

Module 1:

Foundations of Team Science

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Team Science: Why it matters

Data show that
well functioning
teams are more
productive

- 2007 study of 19.9 million papers showed **teams produce higher-impact research**
- Nobel Prize data: 42% of physics **prizes** since 2000 **went to collaborations**
- NIH **success rates higher** for **multi-PI grants** in many programs

Common Pain Points

- Coordination loss
- Uneven contribution
- Group think
- **Unresolved** process conflict
- Goal misalignment

Success Factors: Think **IMPACT**

- **Interdependence:** Members need each other to succeed
- **Motivation:** Shared purpose and individual engagement
- **Processes:** Clear workflows and communication protocols
- **Abilities:** Complementary skills and expertise
- **Culture:** Trust, psychological safety, inclusion norms
- **Tools:** Infrastructure for collaboration and data sharing

Team Science: Reflections

Team Science: Small Group Discussion

- One area where your team was strongest (highest score)
- One area that was most challenging (lowest score)
- **What was it about your collaboration that contributed to those scores?**

*****Please do not name the team or people - focus on the dynamics*****

Team Science: Debrief

- What themes did you hear in your groups about **team strengths**?
- What were **common challenges**?



Team Science: Bookmark this!

- ❖ Clear Shared Goals
- ❖ Complementary Expertise
- ❖ Effective Communication
- ❖ Impartial Participation
- ❖ Conflict Resolution
- ❖ Resource Sharing

Team Science: References for IMPACT

Reference	Team Science Element
National Research Council. (2015). Enhancing the effectiveness of team science. Washington, DC: The National Academies Press. https://doi.org/10.17226/19007	Team processes, Interdependence, Culture and Motivation
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Vogel, A. L., Knebel, A. R., Faupel-Badger, J., Portilla, L. M., & Simeonov, A. (2021). A systems approach to enable effective team science from the internal research program of the National Center for Advancing Translational Sciences. <i>Journal of Clinical and Translational Science</i> , 5(1). https://doi.org/10.1017/cts.2021.811	Processes and Tools
Bennett LM, Gadlin H. Collaboration and Team Science: From Theory to Practice. <i>Journal of Investigative Medicine</i> . 2012;60(5):768-775. https://doi.org/10.2310/JIM.0b013e318250871d	Abilities
National Academies of Sciences, Engineering, and Medicine; Division of Behavioral and Social Sciences and Education; Board on Human-Systems Integration; Board on Behavioral, Cognitive, and Sensory Sciences; Committee on Research and Application in Team Science. (2025). <i>The Science and Practice of Team Science</i> . National Academies Press (US). https://doi.org/10.17226/29043	Team Composition

Team Science: For further reading

Reference

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